

Polyhedral angles and polyhedra

What happens at a vertex of a solid — the angle condition every corner must satisfy.

LESSON 19–20

2 × 40 MIN

GEOMETRY 11, §9

IGCSE E13.X

LESSON CLOCK

12'

22'

22'

20'

18'

6'

What a polyhedral angle is	12 min
The angle inequality	22 min
Convexity	22 min
Euler again	20 min
Practice	18 min
Homework	6 min

LEARNING OBJECTIVES

- Define a polyhedral angle and name its plane and dihedral angles.
- Apply the inequality on the plane angles at a vertex.
- Classify polyhedra as convex or non-convex.
- Use Euler’s formula on a given solid.

TEXTBOOK

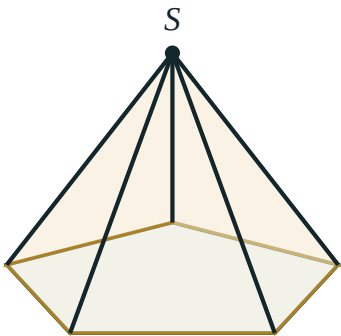
National	pp. 89–100
Cambridge	Core & Extended, pp. 282–288
Workbook	Exercise 9.1

01

Explanation

The corner of a solid

Take a point S and several rays from it, no three coplanar, joined in order by plane faces. The figure is a **polyhedral angle**: three rays give a **trihedral** angle, four a tetrahedral one, and so on.



five plane angles at S , five dihedral angles along the edges
the sum of the plane angles is less than 360°

Five faces meeting at S . Each face contributes a plane angle; each edge a dihedral angle.

TWO KINDS OF ANGLE AT A CORNER

A **plane angle** lies inside one face, between two adjacent edges. A **dihedral angle** is between two adjacent faces, measured along their common edge. A corner with n edges has n of each.

The angle inequality

TWO THEOREMS ABOUT THE PLANE ANGLES

1 Each plane angle is less than the sum of the others. **2** The sum of all the plane angles at a vertex is less than 360° .

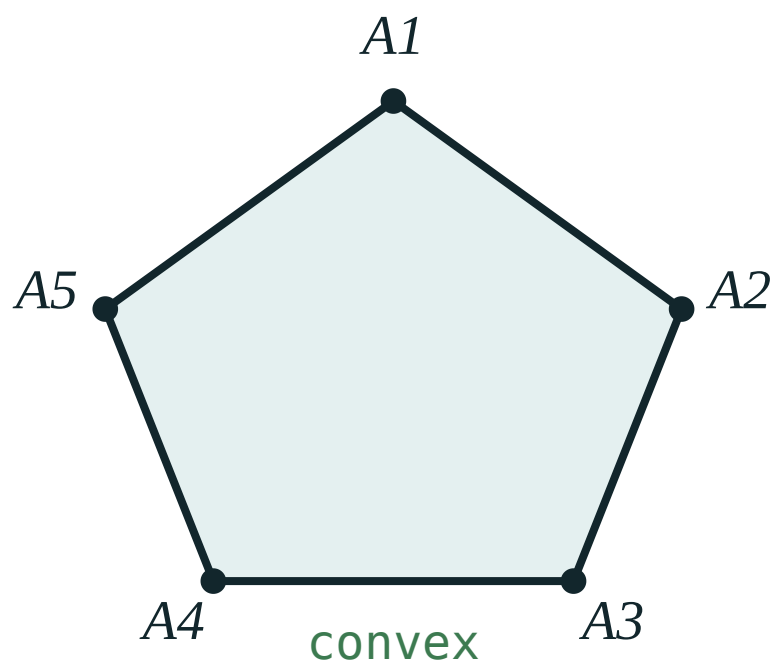
Theorem 2 is the reason there are only five regular polyhedra: with equilateral triangles (60°) at most five fit; with squares (90°) at most three; with regular pentagons (108°) at most three; six hexagons give exactly 360° and lie flat, forming no corner at all.

EXACTLY 360° IS NOT A CORNER

It is a flat piece of plane. To fold up into a solid, a corner must have angles totalling **strictly less** than a full turn — the deficit is what makes it pop out.

Convexity

A polyhedron is **convex** when it lies entirely on one side of the plane of each of its faces — equivalently, when the segment joining any two of its points stays inside.



Convex on the left. On the right, a segment between two interior points leaves the solid.

PROPERTY	CONVEX	NON - CONVEX
Euler's formula $F + V - E = 2$	always	usually, but not for a solid with a hole
every section is a single polygon	yes	no
every plane angle sum $< 360^\circ$	yes	can exceed it at a reflex corner

Euler again, as a tool

$$F + V - E = 2$$

Two counting facts make it usable when the faces are known:

$$2E = \Sigma (\text{edges per face}) = \Sigma (\text{edges per vertex})$$

A solid with 12 pentagons and 20 hexagons, three faces at each vertex: $2E = 12 \times 5 + 20 \times 6 = 180$, so $E = 90$; and $3V = 2E = 180$ gives $V = 60$. Check: $32 + 60 - 90 = 2$
✓. That solid is the football.

02 Worked examples

EXAMPLE 1 Can a corner have three plane angles of 120° each?

- ① $3 \times 120^\circ = 360^\circ$
- ② The sum must be strictly less.
- ③ They lie flat.

ANSWER

No — it is a flat plane, not a corner

EXAMPLE 2 A corner has plane angles 70° , 80° and x . Find the possible values of x .

- ① Sum $< 360^\circ$: $x < 210^\circ$.
- ② Each less than the sum of the others: $x < 150^\circ$.
- ③ And $80 < 70 + x$, so $x > 10^\circ$.

ANSWER

$10^\circ < x < 150^\circ$

EXAMPLE 3 A polyhedron has 12 pentagonal and 20 hexagonal faces, 3 at each vertex. Find V and E .

① $2E = 12 \times 5 + 20 \times 6 = 180$
 $E = 90$

② $3V = 2E = 180$
 $V = 60$

③ Check: $32 + 60 - 90 = 2$ ✓

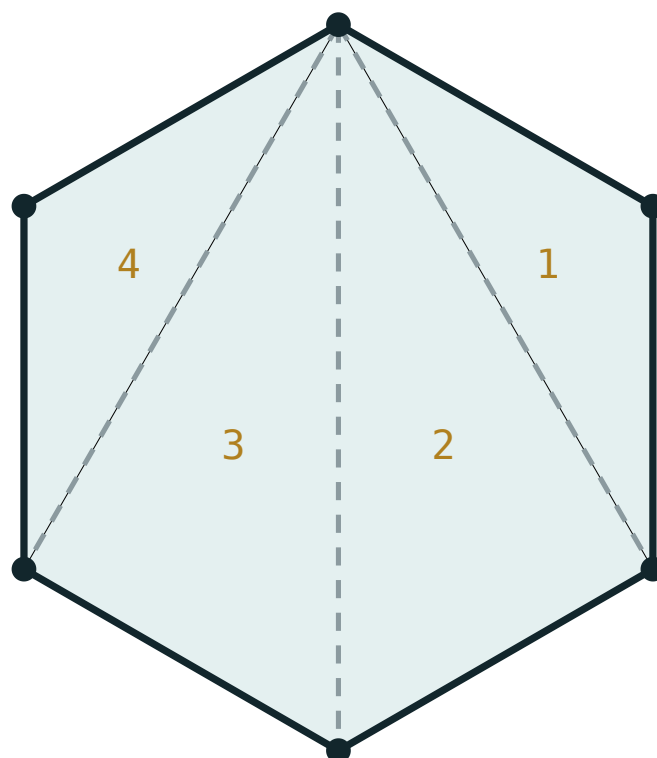
ANSWER

$$V = 60, E = 90$$

03 Interactive model

Fold three squares at a corner and try to add a fourth — the angles refuse.

Angle sum of a polygon



$$180^\circ \times (6 - 2) = 720^\circ$$

sides n **6**

triangles **4**

interior sum **720°**

each angle **120°**

exterior sum **360°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Polyhedral angle	Ko'pyoqli burchak	Многогранный угол
Trihedral angle	Uch yoqli burchak	Трёхгранный угол
Plane angle (of a corner)	Yassi burchak	Плоский угол
Dihedral angle	Ikki yoqli burchak	Двугранный угол
Apex (vertex)	Uchi	Вершина
Convex polyhedron	Qavariq ko'pyoq	Выпуклый многогранник
Non-convex	Qavariq bo'lmagan	Невыпуклый
Euler's formula	Eyler formulasi	Формула Эйлера
Lateral face	Yon yoq	Боковая грань

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The plane angles at a vertex sum to:

A trihedral angle has how many faces?

Three regular hexagons at a corner:

Euler's formula holds for:

every solid

every convex polyhedron

only prisms

only regular solids

$2E$ equals:

the number of faces

the sum of edges per face

$F + V$

$3V$ always

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. How many faces has a trihedral angle?

ANSWER 3

2. Sum of plane angles at a vertex is

ANSWER less than 360°

3. Can three 120° angles form a corner?

ANSWER no

4. Can three 90° angles?

ANSWER yes — a cube corner

5. Can five 60° angles?

ANSWER yes — an icosahedron corner

6. Can six 60° angles?

ANSWER no — exactly 360°

7. State Euler's formula

ANSWER $F + V - E = 2$

Medium · the standard question

7 PROBLEMS

1. Plane angles 70° , 80° , x . Find the range of x

ANSWER $10^\circ < x < 150^\circ$

2. $F = 8$, $V = 6$; find E

ANSWER 12

3. $F = 20$, $E = 30$; find V

ANSWER 12

4. 12 pentagons + 20 hexagons, 3 per vertex: V , E

ANSWER 60, 90

5. A solid with only triangular faces and $V = 6$, 4 per vertex: E , F

ANSWER 12, 8

6. Why are there only five regular polyhedra?

ANSWER The angle sum at a corner must be under 360°

7. Is a picture-frame solid convex?

ANSWER no

Hard · stretch and reason

7 PROBLEMS

1. Prove each plane angle is less than the sum of the others

ANSWER Unfold the corner into the plane and use the triangle inequality

2. Prove the plane angles sum to less than 360°

ANSWER Project the corner onto a plane and compare

3. A convex solid has 7 faces. What is the least possible E ?

ANSWER 12 — a hexagonal pyramid

4. Show $2E \geq 3F$ and $2E \geq 3V$ for any polyhedron

ANSWER At least 3 edges per face and per vertex

5. Deduce $F \leq 2V - 4$ and $V \leq 2F - 4$

ANSWER Substitute both into Euler

6. Can a convex polyhedron have exactly 7 edges?

ANSWER no

7. A solid has faces that are all squares and triangles, 12 vertices, 4 edges at each. Find E , F

ANSWER $E = 24$, $F = 14$

07 Homework

Homework — 5 tasks

Every Euler answer must be checked against $F + V - E = 2$.

1. A corner has plane angles 60° , 100° and x . Find the range of x .
2. Explain, with the angle sums, why there are exactly five regular polyhedra.
3. A solid has 12 faces and 30 edges. Find V and name it if it is regular.
4. A solid has only triangular faces and 20 of them. Find E and V .

5. Give an example of a solid for which Euler's formula fails, and say why.